

Response to Reviewers

Energy-Efficient Deep Learning Without Backpropagation: A Rigorous Hardware-Validated Comparison

Summary

We thank all five reviewers for their time and for a set of comments that turned out to be far more consequential than a normal camera-ready pass. Three of them (R2c, R2d, R3c/R5d) pointed at exactly the places where our evidence was weakest, and acting on them changed our headline result. We would rather report that than defend the accepted version.

The revision rests on a complete re-measurement campaign rather than on rewriting. Concretely:

- **Every experiment was re-run under a harmonised stopping rule.** The accepted version used a *different* early-stopping criterion for each algorithm — a defect our own audit found and no reviewer raised (disclosure D2 below). Because training time and energy are integrals over training duration, an inconsistent stopping rule does not perturb an efficiency comparison, it determines it.
- **The headline efficiency claim does not survive.** Under the harmonised rule MF is $4.0\times$ slower than BP and draws $3.2\times$ the energy on CIFAR-10. The published “41% less energy, 34% faster” *is* reproducible under the submitted protocol — we reproduced it, at 42.4% [38.9, 44.8] and 33.6% [30.2, 36.0] — but its *sign reverses* once the stopping rule is equalised. The abstract, the results and the conclusion have all been rewritten accordingly.
- **What does survive is the accuracy and the memory result.** MF matches or beats BP in every MLP configuration, by 0.01 to 5.04 pp, and holds up to 27% less per-process device memory.
- **Seed counts rose from three to 5 to 122 per contrast**, and every accuracy comparison is now an equivalence test against a preregistered ± 0.25 pp margin with bootstrap confidence intervals.
- **A five-rung ablation ladder from BP to MF** now attributes the cost, including the deep-supervision control that R3 and R5 asked for.
- **We voluntarily disclose five further defects** found by our own audit, none of them raised in review, under *Voluntary disclosures* below. The repository is public and cited in the paper, so all five are discoverable by any reader after publication.

All numbers below are reproducible from the public repository. Section, table and figure references are to the *revised* manuscript, which is **15 pages**, the PPAM limit.

Revised structure. §1 Introduction | §2 Background and Related Work | §3 Framework for Fair and Rigorous Comparison | §4 Experiments and Results (4.1 FF, 4.2 CaFo, 4.3 MF, 4.4 Where the Cost Comes From) | §5 Discussion (5.1 Limitations) | §6 Conclusion. Table 1 = stopping rules; Table 2 = headline summary for all three algorithms; Figure 1 = MF hardware trace; Figure 2 = ablation-ladder waterfall; Figure 3 = equivalence forest.

Reviewer 1

R1a — “Reads like a very condensed technical report ... detrimental to readability for the average PPAM reader.”

Accepted, and this drove a structural rewrite rather than a cosmetic one. The former Sections 2 and 3 were merged into a single §2, Background and Related Work, so that each algorithm is introduced once, in prose, alongside the work it descends from. The contributions list became a paragraph. §4 now leads with the result and follows with the evidence rather than the reverse.

The discussion, which R2 also found thin, was rebuilt around three named questions — why MF is accurate, which memory instrument one should believe, and what the traces imply for hardware co-design — instead of a list of observations.

We also cut hard, because the revision adds a stopping-rule table, an ablation-ladder subsection and an equivalence analysis to a paper that already exceeded the limit. The figure count went from eight to three, the Optuna search-space table and the full environment listing moved to the public repository (cited by URL from the Code and data paragraph), and the three figures we kept were chosen because each carries an argument the prose cannot.

Where: §2 (merged); §4.4; §5; the manuscript is now 15 pages.

R1b — “The plots in Fig. 3 and Fig. 4 are completely illegible.”

Accepted without reservation. Both figures are gone. They were screenshots of a Weights & Biases dashboard, exported at browser resolution, with the dashboard’s own font sizes — unreadable at LNCS column width, and, as it turned out, not reliable either (disclosure D5).

Every figure in the revision is generated from the measurement archive by a committed script, rendered as vector PDF at exactly the LNCS text width (345.827 pt) so that no scaling is applied at inclusion time, with a minimum font size of 8 pt and a colour-blind-safe palette that remains legible in greyscale. Figure 3, the equivalence forest, replaces the old Figure 4 and carries 28 contrasts with their confidence intervals in the space the screenshot used for one panel.

Where: Figures 1–3; `scripts/make_figures.py` in the repository.

Reviewer 2

R2a — “Lack of details and analysis in the main article (specifically Section 6) made the paper seem preliminary.”

Accepted. The discussion was thin because we had description and no attribution. It now has both.

§4.4, *Where the Cost Comes From*, decomposes the BP→MF transition into five single-change rungs: detach the gradient, attach auxiliary supervision, replace the readout, cache activations on device, cache them on host. Measured across all four configurations, **gradient locality alone accounts for 228 to 272 pp of BP’s energy**; auxiliary supervision and the readout together move the total by at most 40 pp and partly cancel on CIFAR; caching contributes nothing. The efficiency result is therefore attributed to a specific architectural property, not to an algorithm taken as a whole.

§5 then explains the mechanism. **No configuration we measured exceeds $\approx 38\%$ mean GPU utilisation**, and MF reaches only 7.3 % against BP’s 16.2 % on CIFAR-10. These workloads are bound by kernel launch latency, not by arithmetic: the layer-sequential schedule issues many small kernels and waits, so the device idles while the run stays resident and the fixed power floor is integrated over four times the duration. That single observation explains the energy result, the absence of any gain from activation caching, and why the conclusion is platform-conditional.

Where: §4.4; Figure 2; §5, all three paragraphs.

R2b — “The details of the different ML architectures are important ... these details are not discussed much in the paper.”

Accepted. §3 now carries a dedicated paragraph, *Architecture, optimiser and implementation*, giving: initialisation (Kaiming uniform for all weights and projection matrices); optimisers (AdamW for BP and BP-DS, Adam on a local cross-entropy loss for CaFo predictors and MF layers); MLP topology (fully connected, ReLU, 2 to 4 hidden layers of 1000 or 2000 units); the exact CaFo block (3×3 convolution, ReLU, 2×2 max-pooling, batch normalisation, 32/128/512 channels, three block

predictors summed); and the FF goodness threshold $\theta = 2.0$ with peer normalisation (factor 0.03, momentum 0.9).

The paragraph also states what we think is the load-bearing implementation fact for a hardware paper: every algorithm is built on the same `nn.Linear` and `nn.Conv2d` primitives in one codebase and differs only in the loss construction and in where `detach()` is called, so no method benefits from a fused or hand-written kernel the others lack. Data augmentation is now stated explicitly and is identical between each algorithm and its baseline.

Where: §3, *Architecture, optimiser and implementation*; §3, *Experimental Setup*.

R2c — “The paper claimed that the BP baselines were identical but algorithm-specific components were removed. The removal of these components may be the reason why certain architectures were better one way or another.”

This is the most valuable comment we received, and it is correct. We did not argue with it; we measured it.

The ablation ladder in §4.4 exists precisely to answer it. It puts the removed components back, one at a time, on top of BP, and measures accuracy and energy at each rung. The result is that the removal is *not* what drives the differences: auxiliary supervision and the projection readout together account for at most 40 pp of the energy change and partly cancel on CIFAR, whereas gradient detachment accounts for 228 to 272 pp.

On accuracy the answer is a split, and we report it as such rather than picking the convenient half. MF minus BP-DS:

Configuration	MF – BP-DS	95% CI	n	Verdict
MNIST 2×1000	+0.11 pp	[−0.01, +0.23]	20	equivalent
Fashion-MNIST 2×1000	−0.06 pp	[−0.17, +0.06]	48	equivalent
CIFAR-10 3×2000	+5.56 pp	[+5.37, +5.74]	20	different
CIFAR-100 3×2000	+6.10 pp	[+5.88, +6.31]	20	different

On the MNIST-scale tasks, auxiliary supervision alone reproduces MF’s accuracy and locality buys nothing. On CIFAR it does not, and MF’s margin is large and fully resolved. A further finding falls out of the same experiment: **BP-DS is worse than plain BP on CIFAR** (−1.65 and −1.06 pp, both resolved), so auxiliary heads are not a free improvement one can bolt onto BP.

Where: §4.4, *Does locality buy anything deep supervision does not?*; Figure 2; Figure 3(a); §3 protocol 2, which now states the objection and points at the measurement.

R2d — “As backprop algorithms are more widely used, the functionalities in existing software libraries could have been better optimised ... than forward-only algorithms.”

The hypothesis was correct in a specific and measurable way, and we tested it rather than arguing about it.

MF’s layer-sequential schedule recomputes the activations of every preceding layer at each stage — an $O(L^2)$ cost that a mature implementation would remove by caching. So we implemented the cache, in two variants (device-resident and host-resident), and verified it against the recomputing path: over 1200 and 3600 optimiser steps the maximum absolute difference in both loss and weights was **exactly zero**. The optimisation is bit-exact.

It bought **no speedup**: 1.00 to 1.02× on device, 0.97× on host, and the device variant costs roughly 40% more per-process memory. The missing optimisation was not missing; it simply does not pay, for the reason given under R2a — at these layer widths the device never exceeds $\approx 38\%$ utilisation and the runs are launch-bound.

The same reading covers the other two algorithms. *Per data pass*, CaFo-Rand is **cheaper** than BP (0.76 to 0.90× Wh/epoch) and FF is only 1.48 to 1.76× more expensive; the end-to-end gaps

are made almost entirely of extra epochs (FF needs 2.66 to $3.82\times$ more, CaFo 3.0 to $7.8\times$ more). Library maturity is not what separates these methods from BP; convergence rate is.

We note that this comment also caused us to *understate* the previously published MF advantage in one respect: the submitted implementation carried the $O(L^2)$ recomputation, so MF’s cost in the accepted version was inflated by an amount we can now bound — and which turns out to be negligible on this hardware.

Where: §4.4, *Is this simply an unoptimised implementation?*; §5, third paragraph; §5.1, limitation 3.

Reviewer 3

R3a — “NVML ... is never formally defined or explained.”

Fixed. NVML is now expanded to the NVIDIA Management Library at first use in the abstract and again at first use in the body, with a statement of what it does and why it was chosen: it reports physical device state — instantaneous board power, SM clock, utilisation, device memory — from the driver itself, so an efficiency claim rests on a measured quantity rather than on an analytical proxy such as a FLOP count.

We went further than the comment asked and **named the API call behind every reported quantity**, because they are not interchangeable: `nvmlDeviceGetPowerUsage` (integrated trapezoidally at 5 Hz for energy), `nvmlDeviceGetUtilizationRates`, `nvmlDeviceGetClockInfo`, `nvmlDeviceGetMemoryInfo`, the host monotonic clock for wall-clock time, `torch.cuda.max_memory_allocated` for per-process memory, and `psutil` for host resident set size. See also disclosure D5.

Where: Abstract; §3, *Evaluation metrics and instruments*; every figure and table caption.

R3b — “The results for CaFo in Section 5.2 are presented purely as text ... should be summarised in an equivalent table.”

Fixed, and generalised. Rather than adding a third standalone table to a paper already over its page limit, Table 2 now presents all three algorithms in one structure: FF, CaFo (both the random-feature and the DFA-pretrained variant) and MF, each against its own tuned BP baseline, with test accuracy, training time, GPU energy and peak per-process memory as mean $\pm$ SD over the stated seed count. CaFo therefore gets the columns the comment asked for, and the reader gets a single point of comparison.

The CaFo numbers are also no longer back-ported from an earlier cluster software stack; all five seeds were re-measured on the current one.

Where: Table 2; §4.2.

R3c — “Because Mono-Forward places a cross-entropy loss at every hidden layer, it inherently behaves like a deep supervision network ... add a standard backpropagation baseline that uses matching layer-wise auxiliary losses. This is the only way to prove whether MF’s gains stem from its unique forward-only mechanism or simply from localised objective constraints.”

Done, and the answer is a split result that we report in full. A BP-DS control was built: identical topology, identical auxiliary cross-entropy head at every hidden layer, identical optimiser and tuning budget — the sole difference being that gradients flow end-to-end instead of being detached at each stage. The table under R2c gives the outcome.

The honest reading, which is now the reading in the abstract: *deep supervision alone reproduces MF’s accuracy on the MNIST-scale tasks, but not on CIFAR, where MF leads by more than 5 pp.* Localised objectives are a sufficient explanation on easy tasks; detachment is doing real work once

the task is hard enough. We have deliberately not generalised in either direction, and we flag the mechanism (a regularisation hypothesis, grounded in the infinite-width analysis of local losses) as a hypothesis, naming the diagnostics that would test it and that we have not run.

We are grateful for this comment in particular: it is the one that forced the abstract to be reframed from “locality causes the saving” to an attributed decomposition.

Where: §4.4, *Does locality buy anything deep supervision does not?*; Figure 3(a), which carries all sixteen ladder contrasts with their intervals; §5, *Why MF is accurate*; abstract.

Reviewer 4

R4a — “The key term ‘hardware-validated comparison’ is unclear and should be explained in more detail.”

Fixed. §3 now opens with a definition before anything else is said: a hardware-validated comparison is one in which *every efficiency claim is backed by a physical quantity read from the GPU itself while the workload runs — energy, elapsed time and memory occupancy — rather than by an analytical proxy such as a FLOP or parameter count.*

The operational content of that definition is supplied immediately afterwards in the instruments paragraph (see R3a), and the specific platform is now stated with the driver version, because the driver governs NVML power-reading semantics: NVIDIA A100-SXM4-40GB on AMD EPYC 7742 nodes, driver 595.71.05, Python 3.10.4, PyTorch 2.4.0+cu121.

Where: §3, opening paragraph; §3, *Experimental Setup* and *Evaluation metrics and instruments.*

R4b — “The statistical analysis is based on only three runs, which is insufficient and can be considered only as preliminary evidence.”

Accepted; this is now closed by a wide margin. Seed counts per contrast are FF $n = 7$, CaFo $n = 5$, MF ladder rungs $n = 20$, and the BP/BP-DS ladder contrasts $n = 60$ (MNIST) and $n = 122$ (Fashion-MNIST) — 5 to 122 rather than three.

We also changed the inferential frame, because more seeds without a decision rule is just a bigger table. Every accuracy comparison is now an **equivalence test** against a margin declared before the runs (± 0.25 pp), with 10 000-sample paired bootstrap confidence intervals. Of 28 contrasts, **eleven are equivalent, sixteen are different, and one is inconclusive**; we report the inconclusive one as inconclusive rather than as evidence of parity. All 28 appear with their intervals in Figure 3.

We additionally report the achieved resolution so a reader can judge what the design can and cannot detect: pooled seed-level SD of test accuracy is 0.284 pp (up to 0.43 pp on the 3×2000 MLPs), giving a bootstrap CI half-width near ± 0.35 pp at $n = 5$ and ± 0.26 pp at $n = 7$.

Where: §3 protocol 5; §4.4, *Intervals, not verdicts.*; Figure 3; §5.1, limitation 4; seed counts in every table and caption.

Reviewer 5

R5a — “The contribution is mainly evaluative rather than algorithmic.”

A fair characterisation of the accepted version, and we have both sharpened what the evaluative contribution is and identified the one place where the work is genuinely structural rather than descriptive.

§4.4 now presents the **time–memory frontier as a consequence of gradient locality, not as an implementation note**. BP, BP-DS and MF-Joint *cannot* cache activations at all: joint gradients make any cached activation stale within one optimiser step. MF can, because detachment plus layer-sequential training freeze every preceding layer for the duration of a stage. The trade-off

axis — spend memory to avoid recomputation — therefore *exists only for the local method*. That it turns out to be a bad trade on this hardware is the empirical part; that it is available at all is the algorithmic part, and it is a property of the objective structure rather than of our code.

We have also stated the scope of the contribution plainly in §5.1 rather than leaving it implied: the framework, the fairness protocols, the ablation ladder and the hardware-level measurements are ours; the three algorithms are not.

Where: §4.4, *The time-memory frontier is a structural property.*; §5, third paragraph; §5.1, limitation 7.

R5b — “Small datasets, shallow architectures, and the use of flattened CIFAR inputs for MLP experiments.”

Accepted and retained as an explicit limitation rather than softened. §5.1 now states that results apply to 2 to 4-layer MLPs and a 3-block CNN, that deeper CNNs, residual networks and transformers are untested, and — the part we think matters most — that flattening CIFAR to 1-D vectors destroys spatial structure and caps absolute accuracy far below CNN state of the art, so *the comparisons are internally valid but say nothing about either algorithm in a competitive vision pipeline*. We flatten because the MF source protocol does; we say so, and we do not claim more.

Where: §5.1, limitations 1 and 2; §3 protocol 1.

R5c — “Results are based on only three runs, and some of the reported accuracy differences are modest.”

The first half is answered under R4b. The second half is the sharper point and we have acted on it directly: modest differences are now *adjudicated* instead of being reported and left to the reader.

Several differences we previously reported as findings do not survive that treatment. On MNIST 3×1000 with SGD the FF-BP accuracy gap is -0.014 pp $[-0.193, +0.144]$ and on MNIST 4×2000 it is -0.064 pp $[-0.219, +0.070]$; both are underpowered and are labelled as such. CaFo-DFA versus BP on the Fashion-MNIST CNN spans $[-0.03, +0.61]$ pp at five seeds and is reported **inconclusive**, not as parity. Conversely, the differences that do matter are now unambiguous: MF over BP by 3.90 and 5.04 pp on CIFAR-10 and CIFAR-100, with intervals nowhere near zero.

Where: §4.1; §4.2; §4.4, *Intervals, not verdicts.*; Figure 3; §5.1, limitation 4.

R5d — “A stronger comparison would include additional baselines, such as deep-supervision or local-loss backpropagation variants, to better isolate whether the gains come from the forward-only training mechanism or from local auxiliary objectives.”

Done — this and R3c are the same request and are answered by the same experiment. Beyond the BP-DS control described under R3c, the ladder adds **MF-Joint** (the MF architecture, including its projection readout, trained with end-to-end gradients), which isolates the readout from the detachment, plus two activation-caching variants. Five rungs in total, so that no single rung conflates two changes.

The isolation the comment asked for is therefore complete: auxiliary objectives, the projection readout, gradient detachment and the caching strategy are each varied alone. The answer, again, is that detachment is what costs and — on CIFAR only — what pays.

Where: §4.4; Figure 2(b), which shows the per-step contribution of all five rungs on all four configurations.

R5e — “The energy measurements are useful, although specific to one GPU platform.”

Accepted, and we have strengthened the caveat rather than merely repeating it, because our own mechanism explanation makes it sharper than the reviewer’s phrasing. Since the central diagnosis is that these workloads are launch-latency-bound rather than compute-bound, the results are *especially* likely to move on hardware with different kernel-launch and memory-bandwidth characteristics. §5.1 now states that we regard the **direction** of the energy result, not merely its magnitude, as platform-conditional, and the conclusion says the same.

We also disclose a second instrument boundary in the same limitation: NVML reports GPU package power only. Host CPU, DRAM and PCIe energy are not instrumented, so any variant that moves work off the accelerator is flattered by the measurement. This matters most for the host-cache MF strategy, whose true total cost is understated by an unknown amount; we still report it as dominated, since it already loses on the instrumented axis.

Where: §5.1, limitations 3 and 5; §5, third paragraph; §6.

R5f — “Clarify the limitations, and strengthen the discussion of baselines and statistical robustness.”

All three parts are addressed above — limitations under R5b and R5e, baselines under R5d, statistics under R4b and R5c. One structural change is worth noting here: §5.1 was rewritten from scratch. The item claiming that three runs were a limitation is gone because it is resolved; in its place are the achieved equivalence margin and minimum detectable effect, the GPU-only energy accounting gap, and an explicit hyperparameter-search confound (each algorithm was tuned independently under an equal trial budget, so differences may partly reflect better hyperparameter discovery, and the search cost — substantial for FF — is excluded from all reported figures).

Where: §5.1 in full.

Voluntary disclosures: defects found by our own audit

No reviewer raised any of the following. We audited our own configuration files, logs and figures while preparing the revision, and found five defects in the accepted version. The repository is public and cited in the paper, so all five are discoverable by any reader after publication. We would rather state them.

D1 — Figure 3 subfloat labels were swapped (factual error in the accepted PDF)

In the CaFo convergence float of the accepted version, the subcaption “Validation Accuracy” pointed at the image plotting mean validation *loss*, and “Validation Loss” pointed at the accuracy image. A reader following the captions would have read both panels backwards.

The float has been removed entirely (it was one of the illegible dashboard exports of R1b) and CaFo’s results are now in Table 2, generated from the measurement archive. Every remaining caption was checked against the image it labels.

D2 — The stopping rule was not consistent across algorithms

This is the single largest correction in the revision. Protocol 4 of the accepted version claimed “consistent early stopping”. The configuration files say otherwise: BP stopped on validation *loss*; FF on validation *accuracy* with a 0.01 minimum improvement; CaFo and MF on per-predictor and per-layer *local* losses, with per-configuration patiences and epoch caps. Worse, those caps apply per *stage*, not per run, so a nominal cap of 15 epochs on a four-stage MF network permits 60 epochs of training against BP’s 100, while BP’s patience was in places as low as 5.

Training time and energy are integrals over training duration. An inconsistent stopping rule therefore does not perturb an efficiency comparison — it determines it, and it is why the 41% energy saving reverses sign once the rule is harmonised. Table 1 gives the submitted settings against the harmonised ones, transcribed from the configuration files, which are preserved in the repository.

D3 — Optuna pruning was asymmetric, in our own favour

BP’s Optuna studies used a median pruner; FF, CaFo and MF ran unpruned. BP’s nominal 50 trials therefore included truncated ones, and its effective search was the weaker of the two. We stress the direction: **the asymmetry disfavoured BP, i.e. it worked in the direction of the accepted paper’s conclusion.** Pruning is now disabled for every algorithm, paid for by a uniform 20-epoch trial cap.

D4 — Three configurations were not independently searched

The Optuna sampler was seeded from the global experiment seed, so two studies over the same space returned byte-identical hyperparameters. The BP MNIST 4×2000 baseline shared its values with BP MNIST 3×1000 , and both CaFo CIFAR-100 variants with their CIFAR-10 counterparts — three configurations whose hyperparameters were in effect transplanted rather than searched. The sampler seed is now derived from the study name; all three were re-searched, and the replacements differ by $2.9\times$, $2.2\times$ and $15.3\times$ in at least one value. A regression test now fails the build if any two configurations share a tuned value.

D5 — Instrument misattribution, and two claims withdrawn

Several efficiency figures in the accepted version were read off Weights & Biases dashboard exports rather than from the measurement archive, and the captions did not name the instrument behind each quantity. Two substantive consequences:

- **Memory was reported from the wrong counter.** The $\approx 5\%$ MF memory saving came from the device-wide NVML counter, in which a fixed ≈ 800 MiB CUDA context — which no algorithm controls — dilutes every difference. Measured per process with `torch.cuda.max_memory_allocated`, the saving is 27% on CIFAR-10 and 25% on CIFAR-100. The correction runs in our favour, which is not a reason to leave it unstated.
- **Two claims are withdrawn.** The accepted version attributed FF’s slowdown in part to “volatile GPU clock speeds ... indicating inefficient hardware saturation”. Both BP and FF hold the SM clock near 1280 MHz; throttling is ruled out, not demonstrated, and the claim is gone. Likewise MF’s “reduced thermal profile” — a 3°C spread on a 23°C to 30°C axis — was never evidence, and that panel has been removed.

Two further corrections of the same kind: the accepted version’s FF slowdown of “up to $13\times$ ” is $7.41\times$ when re-measured under the harmonised rule, and the statement that MF “converges to a lower final validation loss” has been deleted because **no such series was ever recorded** — MF logs only its layer-local objective, which is not the network’s validation loss.

Every efficiency figure in the revision is now traced to a named instrument in §3 and regenerated from the committed measurement archive by a committed script.

Where the revision contradicts the accepted version

For the record, and so that nothing here has to be inferred from the text:

1. **Energy and time.** Accepted: MF uses 41% less energy and trains 34% faster than BP. Revised: under a harmonised stopping rule MF is $4.0\times$ slower and uses $3.2\times$ the energy. The published figures reproduce under the published protocol; the sign reverses under a fair one.
2. **Memory.** Accepted: $\approx 5\%$ saving. Revised: 27% saving, measured per process.

3. **FF slowdown.** Accepted: up to $13\times$. Revised: $7.41\times$.
4. **FF clock volatility and MF thermal profile.** Withdrawn.
5. **MF validation loss curve.** Withdrawn; the series does not exist.
6. **MF accuracy edge on CIFAR-10 under the legacy protocol.** Accepted: +1.21 pp.
Re-measured at three seeds: -0.07 pp $[-0.27, +0.24]$ — the three-seed design could not resolve it. Under the harmonised protocol at twenty seeds the edge is real and larger (+3.90 pp).

The accuracy result and the memory result stand, and the memory result is stronger than we claimed. The efficiency result does not stand, and we say so in the abstract, in §4.3 and in the conclusion.

We thank the reviewers again. The paper is a materially different and, we believe, a more useful one than the version they read.